

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

ANIMAL KINGDOM

LECTURE-2

ZOOLOGY

DAVOOD SIR (CEO
of NG)



Edit with WPS Office

Topics To Be Covered

1

Basis of Classification

②

Phylum Porifera.

2

Previous Year NEET MCQ's

Basis of Classification

Despite the differences in structure and appearance among animals, there are some fundamental features that are common to many of them. These features form the basis for classifying animals, and include:

Level of Organisation.

Body symmetry.

Number of Germ Layers.

Nature of coelom (body cavity)

Notochord.

Segmentation

Digestive system

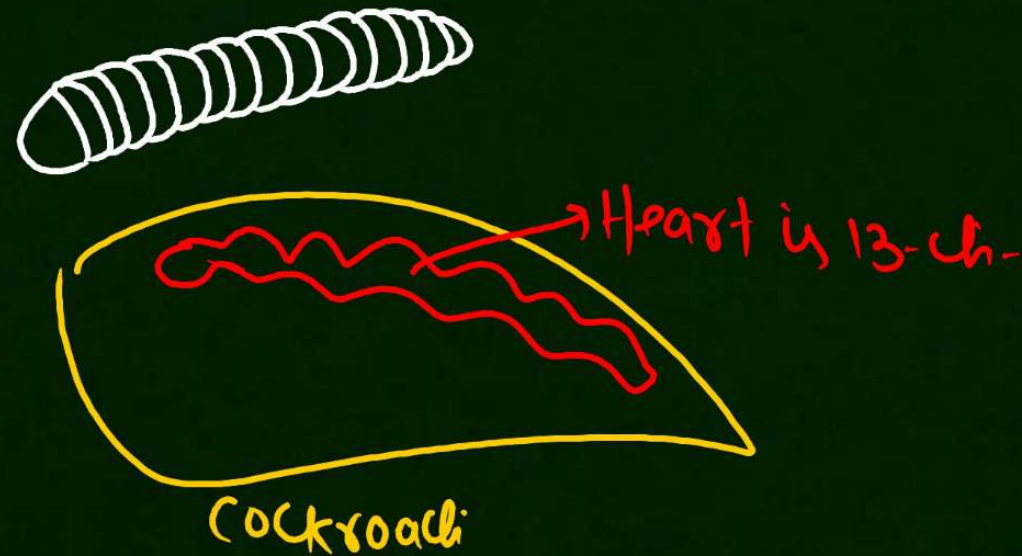
Circulatory system

Body Plan.

These basic characteristics help in grouping animals into different phyla.

5. Segmentation or Metamerism

In some animals, the **body is externally and internally divided into segments** with a serial repetition of at least some organs. For example, in earthworm, the body shows this pattern called **metameric segmentation** and the **phenomenon is known as metamerism**.

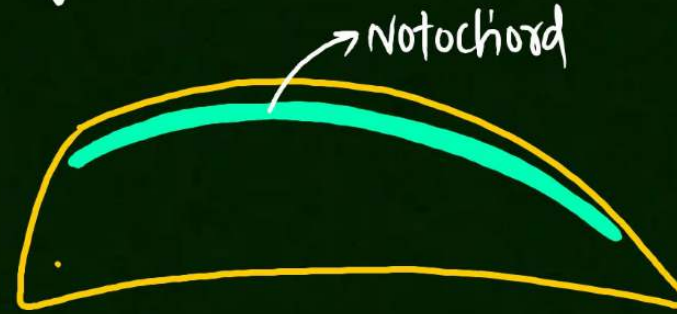


6. Notochord

↳ It is a rod like structure formed dorsally in third week of Embryonic stage. It helps in support.

1. Chordates.

- Animals having notochord at some stage of life.
- Examples: → Fish, Amphibians, Reptiles, Mammals, Birds.



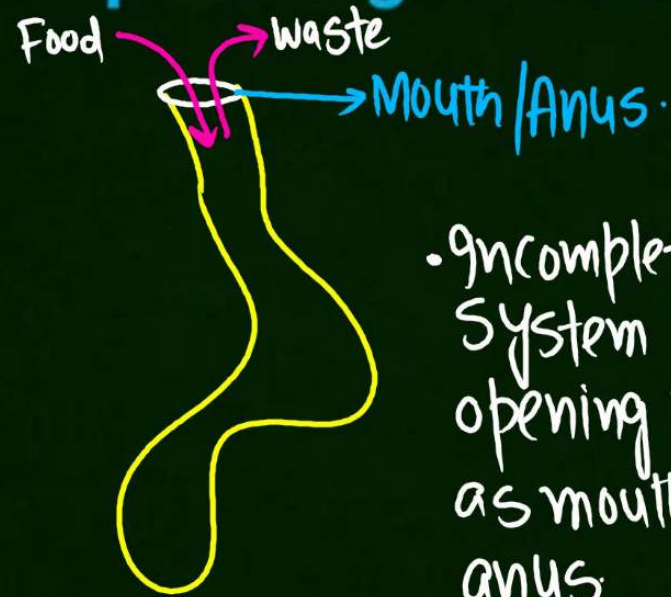
2. Non-Chordates.

- Animals that don't form notochord
- Example: → Porifera to Echinodermata.

7. Digestive System

Digestive System Types

(i) Incomplete Digestive System



- Incomplete digestive system has single opening which acts as mouth as well as anus.
- Example: → platyhelminthes.

(ii) Complete Digestive System

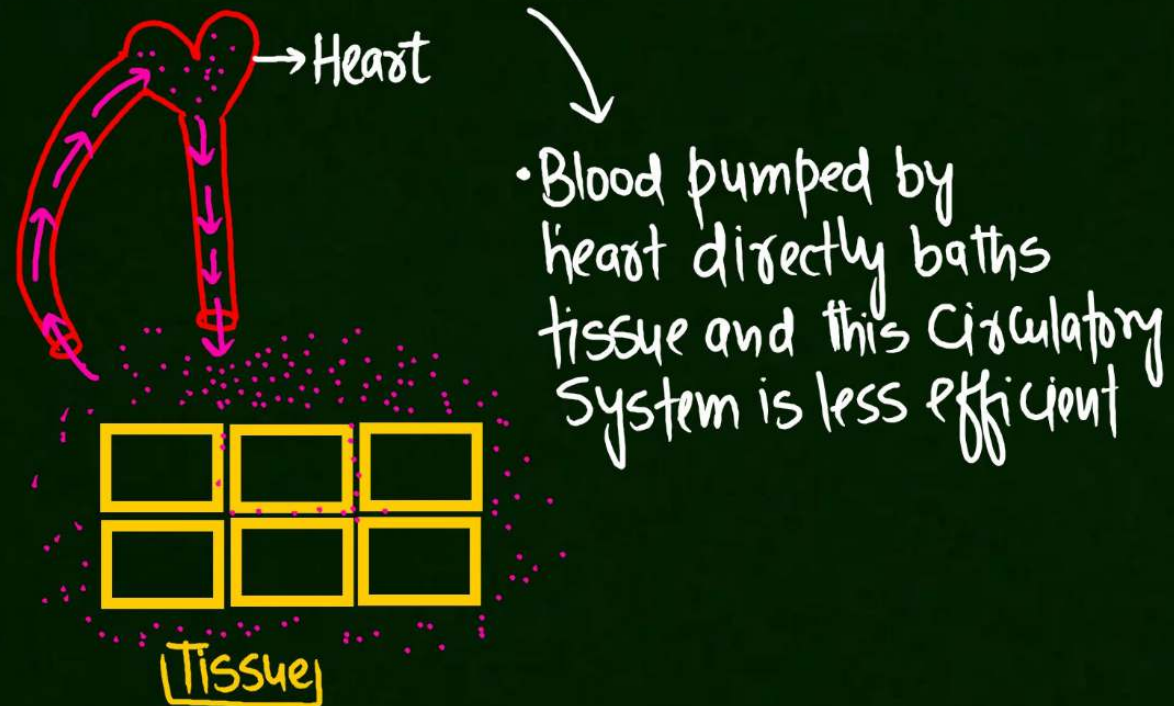


- Complete digestive system has separate opening for intake of food and egestion of food i.e. mouth for ingestion and anus for egestion.
- Example: → chordates.

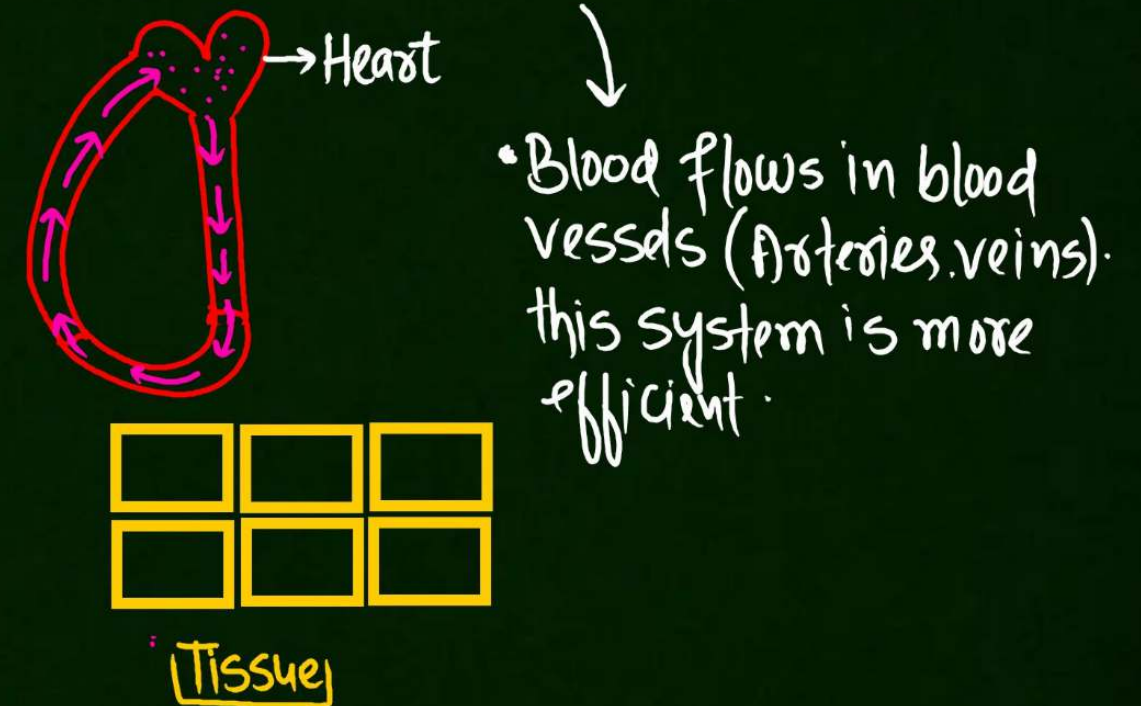
8. Circulatory system

Circulatory System Types

(i) Open Circulatory System

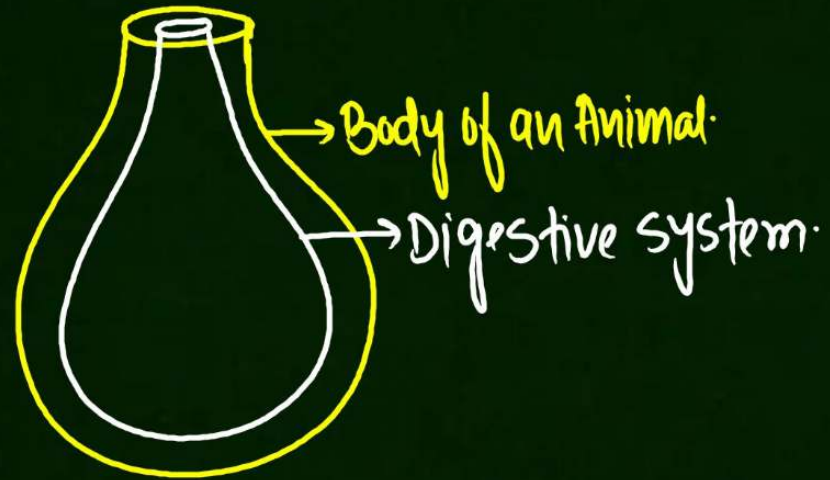


(ii) Closed Circulatory System



9. Body Plan

(i) Blind Sac plan



- A body plan in which digestive system has only one opening that acts as mouth and anus. Food enters and undigested food is eliminated from the same opening because another side of body is blind. Example: Hydra

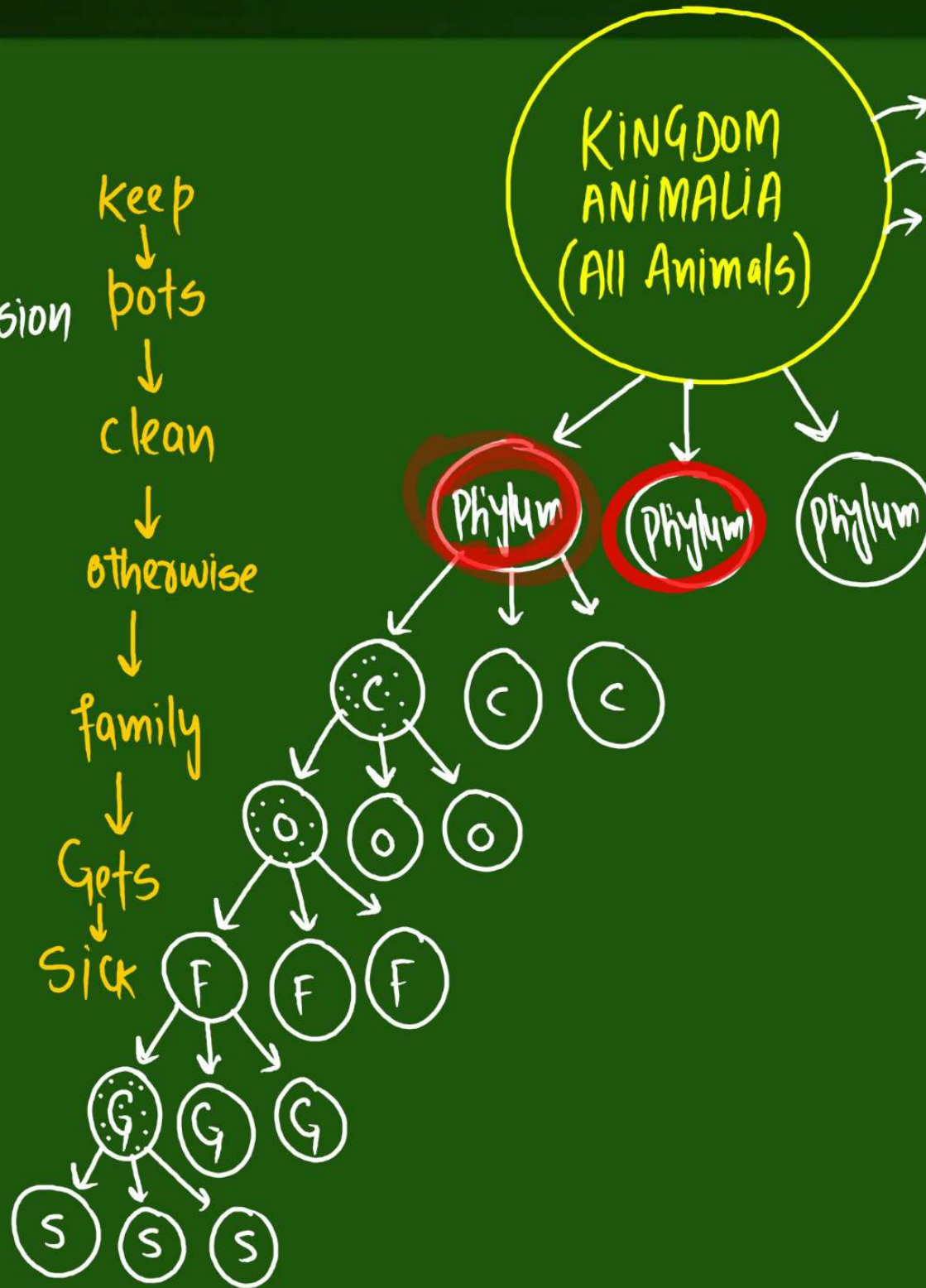
(ii) Tube within a Tube



- A body plan having complete two separate tubes and having mouth and Anus. Food enters through mouth and waste is eliminated through anus.
Example: → Earthworm.

KINGDOM
 ↓
 Phylum/Division
 ↓
 CLASS.
 ↓
 Order
 ↓
 Family
 ↓
 Genus
 ↓
 Species.

Keep
 ↓
 pots
 ↓
 clean
 ↓
 otherwise
 ↓
 family
 ↓
 Gets
 Sick



→ Heterotrophs → Can't make their own food
 → Multicellular → Made of many cells
 → Eukaryotic → Having well defined Nucleus in cells

Phylum-Porifera ^(pore) 9)



Scypha

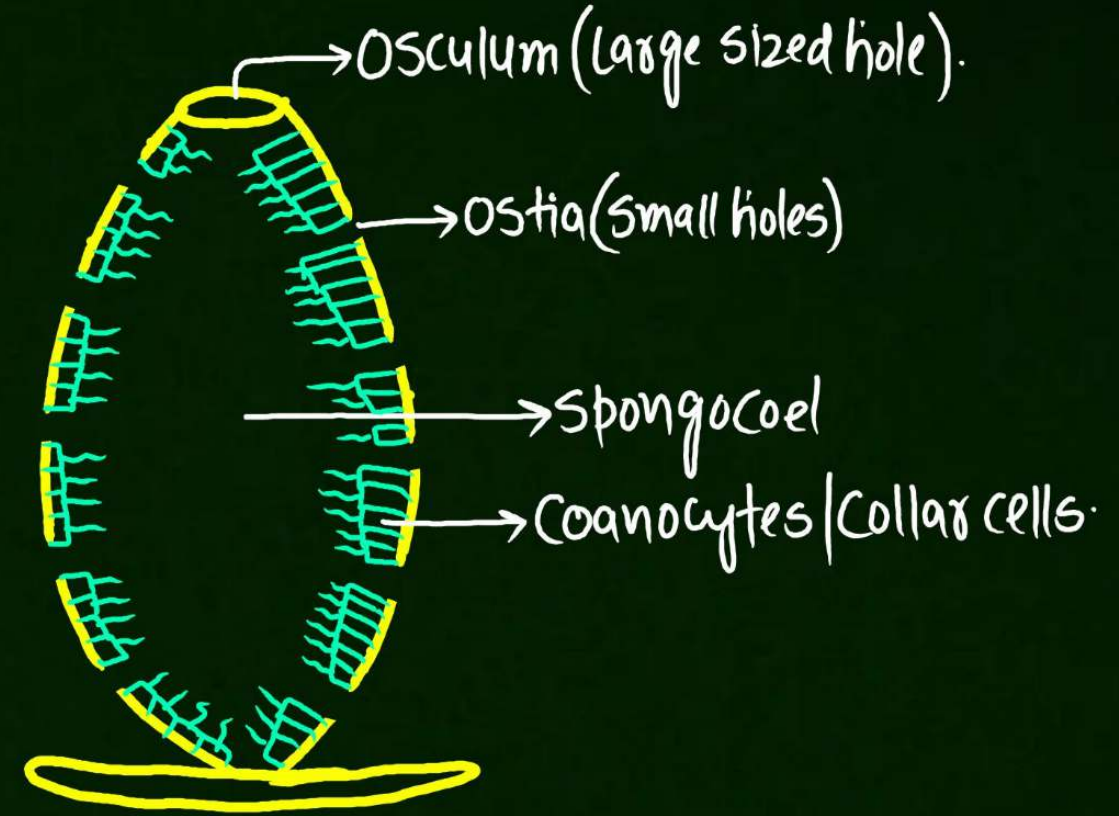


Euspongia



Spongilla

Examples of porifera



Phylum - Porifera (Sponges)

Habitat: → Mostly marine but some are freshwater animals

Symmetry: Usually Asymmetrical.

Organization Level: Cellular level of organization.

Multicellularity: they are primitive (first) multicellular animals

Key Features:

Canal System: water enters through Ostia (tiny pore) flows into Spongocoel (central cavity) and H₂O exits through Osculum (large sized hole). This canal system helps in collection of food, and removal of waste.

Choanocytes (Collar Cells): these cells line Spongocoel. They have flagella which help in capturing of food and movement of H₂O towards Osculum.

Digestion: Intracellular digestion (inside a cell).

Skeleton: Made of spicules (needle like structures) or spongin fibres (protein)

Phylum–Porifera

Reproduction:

Hermaphrodite: → Bisexual, means both the sexes are present in one animal.

Asexual reproduction: Some poriferans reproduce by fragmentation.

Sexual reproduction: → By gamete formation, internal fertilization takes place and indirect development through larval stage.

Examples:

Sycon (Scypha)

Spongilla (freshwater sponge)

Euspongia (bath sponge)

Phylum – Porifera



Phylum–Porifera (Sponges)

"SCARED SIP"

- S** – Sponges (Common name; aquatic, mostly marine)
- C** – Canal system (Ostia → Spongocoel → Osculum)
- A** – Asymmetrical and Cellular level of organization
- R** – Reproduction (Asexual – fragmentation, Sexual – hermaphrodite)
- E** – Endoskeleton made of Spicules or Spongin fibres
- D** – Development indirect (includes larval stage)
- S** – Spongocoel lined with Choanocytes (collar cells)
- I** – Intracellular digestion
- P** – Popular Examples:
 - Sycon (Scypha)
 - Spongilla
 - Euspongia

Kaun Banayga Nabkaar NEET-2017

In case of poriferans, the spongocoel is lined with flagellated cells called

A Oscula

B Choanocytes ✓

C Mesenchymal cells

D Ostia



**Thank
You**